IMPERIAL

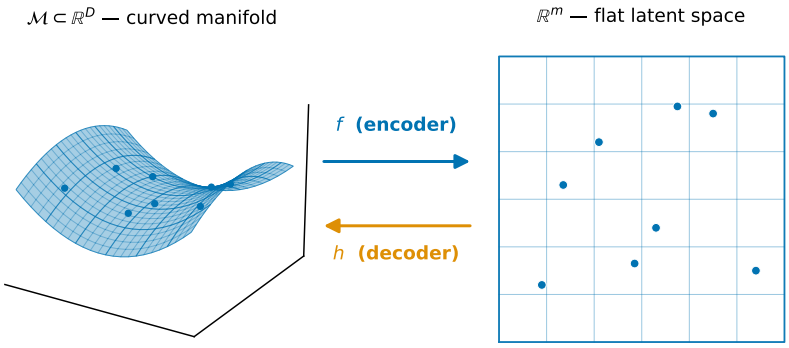
Scale-preserving nonlinear dimension reduction

Homogeneous Autoencoders

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Autoencoders



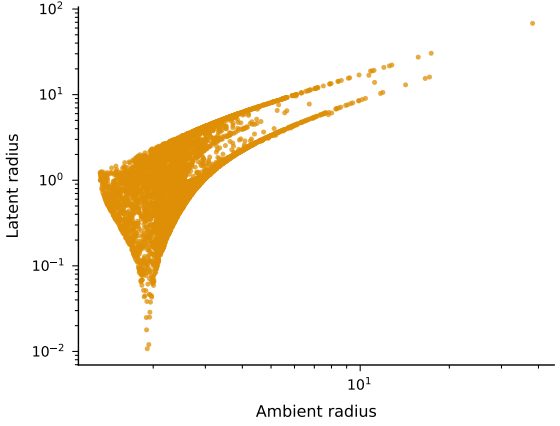
$$X \in \mathbb{R}^D \longmapsto Z = f(X) \in \mathbb{R}^m \longmapsto \widehat{X} = h(Z).$$

Maps f and h parameterised as neural networks and typically trained on $\|X - \widehat{X}\|^2$.

Standard autoencoders distort the magnitude of data

Large $\|X\|$ typically carry the risk and scientific interest. Under a standard AE:

- large $\|X\|$ do not necessarily map to large $\|Z\|$
- latent tail index α_{lat} drifts from the ambient α_{amb}
- yet bulk reconstruction MSE looks fine



Latent vs. ambient radius, standard AE on synthetic data.

Encode the angle, leave the radius untouched

Decompose each observation in polar form: $x = r(x)\,\theta$.

Construction (homogeneous encoder)

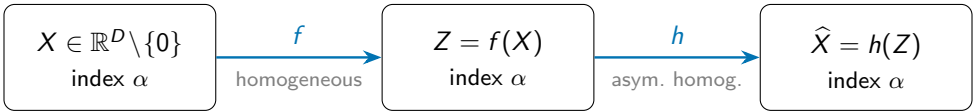
$$f(x) = \underbrace{r(x)}_{\text{radial}}\ \underbrace{e(\theta)}_{\text{angular}}$$

with $e : \mathbb{S}^{D-1} \to \mathbb{S}^{m-1}$ a spherical map parameterised by a neural network.

Radius preserved exactly: $r(f(x)) = r(x)$.

Every radial property of X transports to $Z = f(X)$ analytically.
The decoder, h, mirrors this. Details of the construction are on the poster.

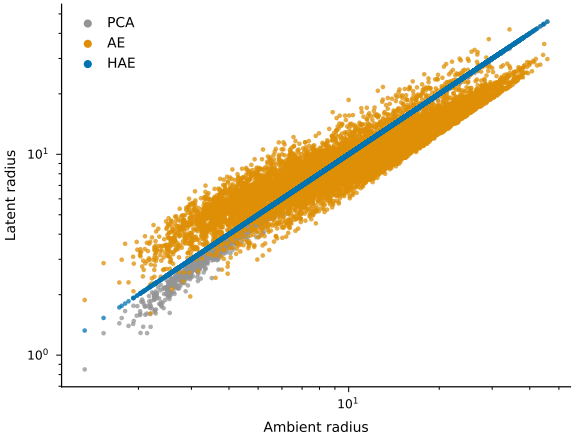
Observation $\to$ latent $\to$ reconstruction round-trip with the same tail index



Extremes stay identifiable in the latent space

Come and chat to me for more details!

HAE preserves the magnitude of data at matched reconstruction cost



ERA5 u_{10} wind over SW England and Wales.
D=195 grid cells, m=8.

	Avg. per-sample MSE
HAE	9.6×10^{-3}
AE	8.9×10^{-3}
PCA	1.4×10^{-2}

- Magnitude of data preserved under HAE.
- Latent radii under standard AE drift from their ambient values into an unidentifiable cloud.
- HAE reconstruction on par with standard AE.

Come to my poster!

Scale-Preserving Autoencoders for High-Dimensional Heavy-Tailed Data

(listed on conference schedule as: Homogeneous Encodings for High-dimensional Extremes)

Why you should look at the poster:

- Nice pictures!
- Wind-field reconstructions, synthetic data manifold plots, Hill-index estimates ...
- Full architecture: encoder and decoder (construction differs slightly).